

Forest Fire Analysis: Predicted Burned Area in Northeast Portugal

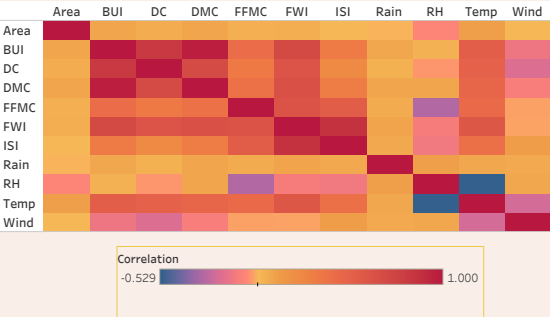
Phase 2 TEPP group project by Amir Benston, Lofinda B. Beynis and Palo Becerra, August 2026.

The purpose of this study was to build a model that could predict what caused some fires to burn more area than others.

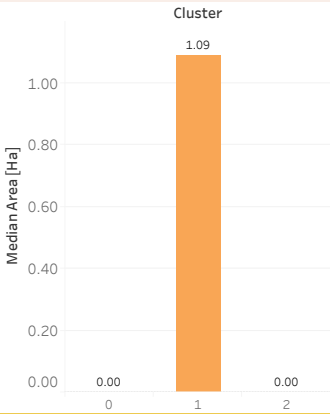
The FWI system rating was the best metric for predicting burned area, beating out the raw feature variables. Even after applying clusters (k=3) to our best model(Model G), we were only able to create a weak regression model.

Sources:
<https://archive.ics.uci.edu/dataset/162/forest+fires>
<https://github.com/lofindanyc/Forest-Fire-Analysis/tree/main>

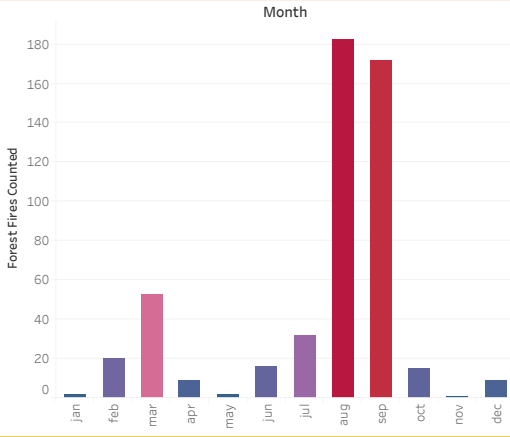
Correlation Heatmap: No Strong Predictors Re: Fire Size
No single weather variable or index correlated strongly with burned area.
The strongest was temp at a mere 0.10.



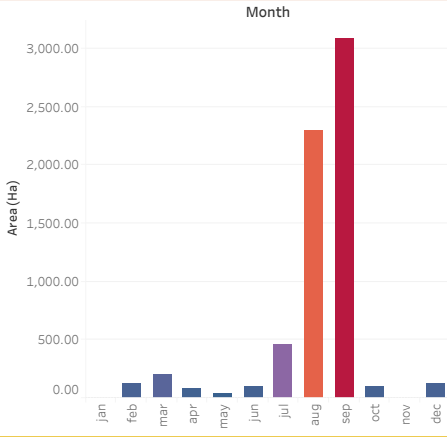
One Cluster Stood Alone:
Fires' Habitual Spread
Cluster 1 was the only group where measurable fire spread was the typical outcome.



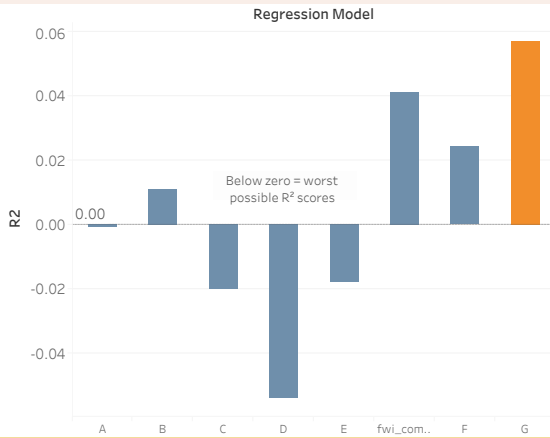
Forest Fires by Month: August + September Have Most Activity
Late summer weather tends to be hot and dry in the mountainous region of Northeast Portugal.



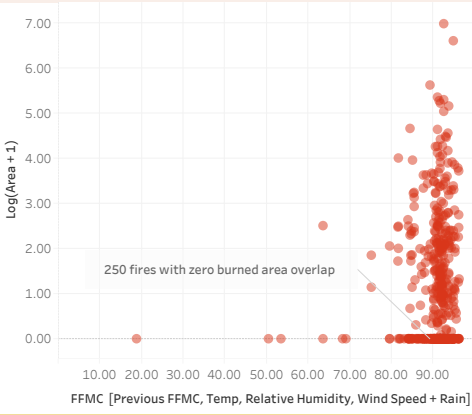
Burned Area by Month: September Peaks
Despite having fewer forest fires than August did, September consistently rose ahead as having the most burned area (3,086).



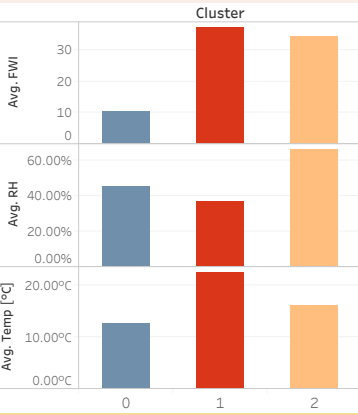
Eight Models, One Modest Win: Model G
No single variable, index, or model reliably predicted fire size.
Grouped risk clusters came closest.



FFMC vs Log(Area + 1):
Standout Predictor Was FFMC Index
Amongst all the raw variables, a composite feature was the only meaningful clue for further analysis.



K-3 Clusters:
Subtle Signals to Prediction
Patterns key to usage in Model G - the best performing model during analysis.



Predicted vs Actual: Model G
With such a difficult dataset, even the best model underpredicted the largest of forest fires.

